looking for alternatives to RDBMS. This led to the growth of non-relational distributed databases. These non-relational distributed database systems, which vary widely in their design, have come to be referred to by the term 'NoSQL', signifying that they are different from the traditional relational database systems that support a structured query language.

## What is NoSQL?

NoSQL stands for 'Not-Only-SQL'. It is the emergence of a growing number of non-relational, distributed data stores that typically do not attempt to provide ACID guarantees. NoSQL databases may not require fixed table schemas, and they typically scale horizontally. NoSQL architecture often provides weak consistency guarantees and restricted transactional support.

The origins of the NoSQL movement can be traced to what is now known as CAP theorem (*http://en.wikipedia.org/wiki/CAP_theorem*). Recall the fact that in order to handle massive datasets, databases turned to 'Shared Nothing' partitioned systems. In 2002, Eric Brewster made the following conjecture in the PODC conference keynote talk (*http:/www.cs.berkeley.edu/~brewer/cs262b-2004/PODC-keynote.pdf*):

"It is impossible for a distributed computer system to simultaneously provide all three of the following guarantees:

1. Consistency (all nodes see the same data at the same time)
2. Availability (node failures do not prevent survivors from continuing to operate)
3. Partition tolerance (the system continues to operate despite arbitrary message loss)"

According to the theorem, a distributed system can satisfy any two of these guarantees at the same time, but not all three. Therefore, the 'Shared Nothing' partitioned database architectures moved towards non-relational databases, which sacrificed consistency in order to provide high availability, scalability and partition tolerance. These partitioned non-relational databases moved from the traditional ACID semantics supported by RDBMS to BASE semantics. BASE stands for 'Basically Available, Soft state, Eventually consistent'. Under the BASE semantics, it is enough for the database to eventually be in a consistent state. ACID is pessimistic, and forces consistency at the end of every transaction. BASE is optimistic, and accepts that the database consistency will be in a state of flux. In simple terms, eventual consistency means that while a database may be inconsistent at certain points of time, it will eventually become consistent; i.e., eventually, all database nodes will receive the latest consistent updates. This relaxed consistency allows BASE systems to provide high scalability.

A detailed discussion of ACID versus BASE semantics, and how correctness can still be maintained under BASE semantics, can be found in the ACM queue article available at *http://queue.acm.org/detail.cfm?id=1394128*. Incidentally, it was this article that coined the term 'BASE' for such partitioned database architectures.

## Types of NoSQL systems

Several NoSQL systems employ a distributed architecture, similar to the 'Shared-Nothing' model. Simplistic implementations use associative arrays or key-value pairs. Often, they are implemented with distributed hash table (DHT), multi-dimensional tables, etc. A wide variety of distinct families exist under the NoSQL movement. Let's look at these families now.

### Simple key-value store

This is a schema-less data store that allows the application to store its data, and allows it to manage, most often with minimal data-type support. Dynamo (Amazon), Voldemort (LinkedIn) use the Key-Value Distributed Hash Table from implementing their databases. SimpleDB provides simple key/value pairs, stored in a distributed hash table, but interoperates well with the rest of its cloud building blocks. Dynamo is Amazon's major research paradigm for non-relational database design, with a simpler model with massive scaling abilities, and extraordinarily high availability requirements. It allows applications to relax their consistency guarantees, under certain failure scenarios. The techniques used to make Dynamo work and perform well include consistent hashing, vector clocks, Merkle tree data structure and Gossip (a distributed information sharing approach). Dynomite is an open source implementation of Dynamo, written in Erlang. A detailed description of Dynamo is available from Werner Vogel's paper at *http://www.allthingsdistributed.com/files/amazon-dynamo-sosp2007.pdf.*

### Table-oriented NoSQL data stores

These are similar to key-value stores, but define the value as a set of columns. An example of a table-oriented NoSQL store is Google's BigTable. One of the leaders in this space, Google's BigTable is integrated with its cloud computing platform (Google App Engine) and offers simplified SQL capabilities called GQL. BigTable and its clones are implemented as sparse, multi-dimensional sorted maps, instead of simple key-value stores. The row, column and timestamp are used to index data. Atomicity is guaranteed at a low-level row (updates to a single row are always transactional). Locality groups provide the ability to combine different columns to impose access control and hints about data that are typically accessed together. Hypertable and HBase are two open source clones of BigTable.

One of the key differences between these implementations is the underlying file system that actually stores the data. BigTable uses Google's proprietary file system (GFS) while the open source clones use HadoopFileSystem or KosmosFileSystem (KFS). These are open source implementations of Google File System. HBase (*hbase.apache.org*), in turn, provides additional database capabilities, map/reduce integration, etc.